Chapter 8. Loglinear Models for Contingency Tables

Two-, three- and higher-way tables

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Loglinear models are log-link Poisson GLMs for cell counts that treat all variables symmetrically as joint responses; logit models with categorical predictors are special cases (Section 8.5). $\{n\}$, $\{\mu\}$, $\{\pi\}$ denote observed counts, expected counts and cell probabilities; "+" denotes summation.

8.1 Loglinear models for two-way tables

Motivation. Independence $\pi_{ij} = \pi_{i+}\pi_{+j}$ is additive on the log scale; adding "interaction" terms describes any association.

$\mu_{ij} = n\pi_{ij}$ (multinomial) or $\mu_{ij} = E(Y_{ij})$ for $N = IJ$ independent Poisson counts; formulas in μ_{ij} cover both. Rows are X, columns Y.

8.1.1 Independence model

Definition (Loglinear model of independence)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \tag{8.1}$$

with identifiability constraints such as $\lambda_I^X = \lambda_J^Y = 0$ (baseline coding) or $\sum_i \lambda_i^X = \sum_j \lambda_j^Y = 0$.

Theorem 8.1 (Independence as a loglinear model)

For positive $\{\mu_{ij}\}$ with $\mu_{ij} = n\pi_{ij}$, the variables X and Y are independent if and only if (8.1) holds for some parameters. The ML fitted values are $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$, then $\log \mu_{ij} = \log n + \log \pi_{i+} + \log \pi_{+j}$. Put $\lambda_i^X = \log \pi_{i+} - \log \pi_{I+}$, $\lambda_j^Y = \log \pi_{+j} - \log \pi_{+J}$ and $\lambda = \log n + \log \pi_{I+} + \log \pi_{+J}$; this is (8.1) with baseline constraints.

Step 2. Conversely, if $\mu_{ij} = a_i b_j$ with $a_i = e^{\lambda+\lambda_i^X}$, $b_j = e^{\lambda_j^Y}$, then $\mu_{i+} = a_i b_+$, $\mu_{+j} = a_+ b_j$, $\mu_{++} = a_+ b_+$, so

$$\pi_{ij} = \frac{a_i b_j}{a_+ b_+} = \frac{a_i b_+}{a_+ b_+} \cdot \frac{a_+ b_j}{a_+ b_+} = \pi_{i+}\pi_{+j}.$$

Step 3. The fitted values $n_{i+}n_{+j}/n$ satisfy the model and reproduce the row and column totals; by Birch's result (Theorem 8.19) they are the ML estimates. ∎

Hence the Pearson X^2 and G^2 tests of independence are goodness-of-fit tests of the loglinear model (8.1), with $\mathrm{df} = (I-1)(J-1)$.

8.1.2 Interpretation of parameters

A loglinear GLM takes the N cell counts as observations and does not distinguish response from explanatory variables; forming a logit still helps interpretation.

Proposition 8.2 (Independence means a constant logit)

Let $J = 2$. Under (8.1),

$$\operatorname{logit} P(Y = 1 \mid X = i) = \lambda_1^Y - \lambda_2^Y \qquad \text{for every } i,$$

so the odds of column 1 equal $\exp(\lambda_1^Y - \lambda_2^Y)$ in every row.

Proof. $P(Y=1 \mid X=i)/P(Y=2 \mid X=i) = \mu_{i1}/\mu_{i2}$, hence the logit equals

$$\log \mu_{i1} - \log \mu_{i2} = (\lambda + \lambda_i^X + \lambda_1^Y) - (\lambda + \lambda_i^X + \lambda_2^Y) = \lambda_1^Y - \lambda_2^Y,$$

which does not depend on i. ■

Differences of main-effect parameters of one variable are log odds of one of its categories versus another.

8.1.3 Saturated model

Definition (Saturated two-way model)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}. \tag{8.2}$$

The association terms λ_{ij}^{XY} (coefficients of products of dummy variables) measure departures from independence; with baseline constraints $\lambda_{Ij}^{XY} = \lambda_{iJ}^{XY} = 0$.

Theorem 8.3 (Saturated model and odds ratios)

(a) Model (8.2) has IJ nonredundant parameters and describes every positive table $\{\mu_{ij}\}$ exactly.
(b) Every odds ratio is a contrast of the association parameters; for rows a, c and columns b, d,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}.$$

In particular, for a 2×2 table $\log \theta = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}$.
(c) Independence holds iff all $\lambda_{ij}^{XY} = 0$; the test has df $= (I-1)(J-1)$.

Proof.

Step 1. Counting under baseline constraints: $1 + (I-1) + (J-1) + (I-1)(J-1) = IJ$.

Step 2. Given positive $\{\mu_{ij}\}$, set $h_{ij} = \log \mu_{ij}$ and define

$$\lambda = h_{IJ}, \quad \lambda_i^X = h_{iJ} - h_{IJ}, \quad \lambda_j^Y = h_{Ij} - h_{IJ}, \quad \lambda_{ij}^{XY} = h_{ij} - h_{iJ} - h_{Ij} + h_{IJ}.$$

These satisfy the baseline constraints and $\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY} = h_{ij}$, so every positive table is represented (uniquely, since the map is invertible).

Step 3. For (b), substitute (8.2): in $h_{ab} + h_{cd} - h_{ad} - h_{cb}$ the terms λ, $\lambda_a^X, \lambda_c^X, \lambda_b^Y, \lambda_d^Y$ each appear once with $+$ and once with $-$, leaving the stated contrast.

Step 4. For (c), by step 2 all $\lambda_{ij}^{XY} = 0$ iff $h_{ij} - h_{iJ} - h_{Ij} + h_{IJ} = 0$ for all i, j, i.e. all odds ratios with the last row and column are 1, which is equivalent to independence. The $(I-1)(J-1)$ free association parameters give the df. ■

Hierarchical models

A model is *hierarchical* if each term comes with all lower-order terms in its variables (e.g. λ_{ij}^{XY} requires λ_i^X, λ_j^Y). Nonhierarchical models such as $\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_{ij}^{XY}$ depend on the coding (baseline coding forces $\log \mu_{Ij} = \lambda$) and are rarely sensible. Interpret the highest-order terms, not main effects.

8.1.4 Alternative parameter constraints

Sum-to-zero constraints $\sum_i \lambda_{ij}^{XY} = \sum_j \lambda_{ij}^{XY} = 0$ are an alternative; only contrasts like Theorem 8.3(b) are invariant. For 2×2 with $\log\theta = 2.0$: baseline coding gives $\lambda_{11}^{XY} = 2.0$, others 0; sum-to-zero gives $\lambda_{11}^{XY} = \lambda_{22}^{XY} = 0.5$, $\lambda_{12}^{XY} = \lambda_{21}^{XY} = -0.5$. Baseline coding allows some estimates in a set to be infinite while others stay finite.

8.1.5 Multinomial models for cell probabilities

Proposition 8.4 (Poisson to multinomial)

If Y_{ij} are independent Poisson with $\log\mu_{ij}$ given by (8.2), then conditional on $\sum Y_{ij} = n$ the counts are multinomial$(n, \{\pi_{ij}\})$ with

$$\pi_{ij} = \frac{\exp(\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY})}{\sum_a \sum_b \exp(\lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}^{XY})}, \tag{8.3}$$

in which λ cancels.

Proof. Let $\mu_{++} = \sum \mu_{ij}$; the total is Poisson(μ_{++}). For counts $\{n_{ij}\}$ summing to n,

$$P(\{Y_{ij} = n_{ij}\} \mid Y_{++} = n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!}{e^{-\mu_{++}}\mu_{++}^n/n!} = \frac{n!}{\prod n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu_{++}}\Big)^{n_{ij}},$$

since $\prod e^{-\mu_{ij}} = e^{-\mu_{++}}$ and $\sum n_{ij} = n$. This is multinomial with $\pi_{ij} = \mu_{ij}/\mu_{++}$, which is (8.3); the common factor e^{λ} cancels. ∎

The intercept λ only reflects the total, random under Poisson and fixed under multinomial sampling.

8.2 Loglinear models for independence and interaction in three-way tables

Motivation. Each kind of three-way independence or association is a loglinear model with certain terms absent, so the model symbol shows the structure.

The following lemma turns every independence statement into a statement about which terms appear.

Lemma 8.5 (Generating classes and factorization)

Let a hierarchical loglinear model have generating class $\{C_1, \ldots, C_m\}$ (the highest-order terms, i.e. the model symbol; e.g. (XZ, YZ) has $C_1 = \{X, Z\}$, $C_2 = \{Y, Z\}$). Then a positive table $\{\mu\}$ satisfies the model if and only if

$$\mu(\mathbf{x}) = \prod_{r=1}^{m} \psi_r(\mathbf{x}_{C_r})$$

for some positive functions ψ_r that depend on the cell $\mathbf{x}$ only through the coordinates in C_r.

Proof. [Supplement]

Step 1. Every term of the model involves a subset of some C_r, hence is a function of $\mathbf{x}_{C_r}$. Grouping terms by a generator containing them, $\log\mu = \sum_r g_r(\mathbf{x}_{C_r})$; exponentiate to get the product form.

Step 2. Conversely, by Theorem 8.3 (step 2, extended to more variables) any $g(\mathbf{x}_C)$ is a constant plus main effects plus interactions of every subset of C, all of which are model terms (hierarchy). Summing these decompositions of $\log \psi_r$ gives $\log \mu$ in the model's form. ■

8.2.1 Types of independence

Consider an $I \times J \times K$ table of three responses X, Y, Z with probabilities $\{\pi_{ijk}\}$ (or Poisson means $\{\mu_{ijk}\}$).

Definition (Three kinds of independence)

(i) *Mutual independence*: $\pi_{ijk} = \pi_{i++}\pi_{+j+}\pi_{++k}$ for all i, j, k.
(ii) *Joint independence* of Y from (X, Z): $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$, i.e. two-way independence between Y and the IK combinations of (X, Z).
(iii) *Conditional independence* of X and Y given Z: $\pi_{ij|k} = \pi_{i+|k}\pi_{+j|k}$ for all k, where $\pi_{ij|k} = P(X = i, Y = j \mid Z = k)$; equivalently

$$\pi_{ijk} = \pi_{i+k}\pi_{+jk}/\pi_{++k} \qquad \text{for all } i, j, k. \tag{8.4}$$

Theorem 8.6 (Independence structures as loglinear models)

For positive tables:
(a) mutual independence $\iff$ (X, Y, Z): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$;
(b) Y jointly independent of (X, Z) $\iff$ (XZ, Y): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ}$;
(c) $X \perp Y \mid Z$ $\iff$ (XZ, YZ): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}$.
Moreover (a) $\Rightarrow$ (b) $\Rightarrow$ (c).

Proof. By Lemma 8.5 it suffices to match each independence statement with a product form. Write $\mu = n\pi$.

Step 1 (c). If (8.4) holds, $\mu_{ijk} = \psi_1(i,k)\psi_2(j,k)$ with $\psi_1 = n\pi_{i+k}$ and $\psi_2 = \pi_{+jk}/\pi_{++k}$. Conversely, if $\mu_{ijk} = A_{ik}B_{jk}$, then $\mu_{i+k} = A_{ik}B_{+k}$, $\mu_{+jk} = A_{+k}B_{jk}$ and $\mu_{++k} = A_{+k}B_{+k}$, whence

$$\frac{\mu_{i+k}\mu_{+jk}}{\mu_{++k}} = \frac{A_{ik}B_{+k}\, A_{+k}B_{jk}}{A_{+k}B_{+k}} = A_{ik}B_{jk} = \mu_{ijk},$$

which is (8.4) after dividing by n.

Step 2 (b). If $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$ we have a product $\psi_1(i,k)\psi_2(j)$. Conversely if $\mu_{ijk} = A_{ik}B_j$ then $\mu_{i+k} = A_{ik}B_+$, $\mu_{+j+} = A_{++}B_j$, $\mu_{+++} = A_{++}B_+$, and $\mu_{i+k}\mu_{+j+}/\mu_{+++} = A_{ik}B_j$, i.e. $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$.

Step 3 (a). Same computation with $\mu_{ijk} = A_iB_jC_k$: $\mu_{i++}\mu_{+j+}\mu_{++k}/\mu_{+++}^2 = A_iB_jC_k$.

Step 4. The implications follow because the model in (a) is a special case of (b) (set $\lambda^{XZ} = 0$), which is a special case of (c) (set $\lambda^{YZ} = 0$). ■

Conditional and marginal independence do not imply each other (Simpson's paradox); both hold under the stronger structures (Figure 1).

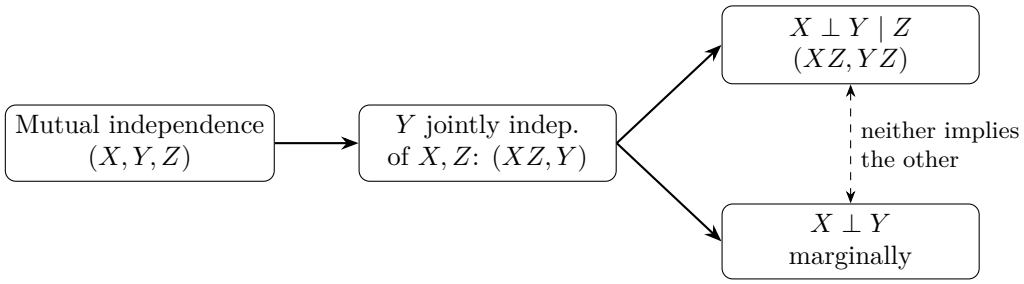


Figure 1: Relationships among types of XY independence (book Figure 8.1).

8.2.2 Homogeneous association and three-factor interaction

Definition (Homogeneous association and the saturated model)

The model of *homogeneous association* (no three-factor interaction) is

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}, \tag{8.5}$$

symbol (XY, XZ, YZ); equivalently $\pi_{ijk} = \psi_{ij}\phi_{jk}\omega_{ik}$. The general (saturated) model adds λ_{ijk}^{XYZ}, symbol (XYZ).

The saturated model has

$$1+(I-1)+(J-1)+(K-1)+(I-1)(J-1)+(I-1)(K-1)+(J-1)(K-1)+(I-1)(J-1)(K-1) = IJK$$

parameters, one per cell (expand $IJK = [(I-1)+1][(J-1)+1][(K-1)+1]$). Unlike the models of Theorem 8.6, (8.5) has no closed-form expression of ψ, ϕ, ω in terms of margins.

Model symbol	Association terms	Interpretation
(X, Y, Z)	none	mutual independence
(XY, Z)	λ^{XY}	Z jointly independent of X, Y
(XY, YZ)	$\lambda^{XY}, \lambda^{YZ}$	$X \perp Z \mid Y$
(XY, XZ, YZ)	all two-factor	homogeneous association
(XYZ)	all, incl. λ^{XYZ}	saturated

8.2.3 Interpreting model parameters

At level k of Z the XY conditional association is described by the $(I-1)(J-1)$ local odds ratios

$$\theta_{ij(k)} = \frac{\mu_{ijk}\,\mu_{i+1,j+1,k}}{\mu_{i,j+1,k}\,\mu_{i+1,j,k}}, \tag{8.6}$$

and similarly $\theta_{i(j)k}$ for XZ and $\theta_{(i)jk}$ for YZ. Conditional independence of X, Y given Z means all $\theta_{ij(k)} = 1$.

Theorem 8.7 (Homogeneous association)

(a) Under (XY, XZ, YZ),

$$\log \theta_{ij(k)} = \lambda_{ij}^{XY} + \lambda_{i+1,j+1}^{XY} - \lambda_{i,j+1}^{XY} - \lambda_{i+1,j}^{XY} \quad \text{for all } k, \tag{8.7}$$

and similarly for XZ and YZ odds ratios.

(b) Conversely, a positive table satisfies (XY, XZ, YZ) if and only if $\theta_{ij(1)} = \cdots = \theta_{ij(K)}$ for all i, j. Equivalently, iff the XZ odds ratios are the same at every j, or iff the YZ odds ratios are the same at every i.

Proof. [Supplement]

Step 1. For (a), substitute (8.5) into $\log \theta_{ij(k)}$. Each of $\lambda, \lambda^X, \lambda^Y, \lambda^Z_k, \lambda^{XZ}_{ik}, \lambda^{XZ}_{i+1,k}, \lambda^{YZ}_{jk}, \lambda^{YZ}_{j+1,k}$ appears once with each sign and cancels; only the XY contrast remains, and it does not involve k.

Step 2. For (b), suppose $\theta_{ij(k)} = \theta_{ij}$ does not depend on k. Let $h_{ijk} = \log \mu_{ijk}$. Telescoping the local log odds ratios over the rectangle $[1, i] \times [1, j]$,

$$h_{ijk} - h_{1jk} - h_{i1k} + h_{11k} = \sum_{a<i} \sum_{b<j} \log \theta_{ab} =: g(i, j),$$

which is free of k. Therefore

$$h_{ijk} = g(i, j) + [h_{1jk} - h_{11k}] + h_{i1k},$$

a sum of a function of (i, j), a function of (j, k) and a function of (i, k). By Lemma 8.5 the table satisfies (XY, XZ, YZ).

Step 3. The symmetric statements follow by permuting the variables; each is equivalent to (XY, XZ, YZ). ■

The three-factor term describes how an odds ratio changes across the third variable. For $2 \times 2 \times 2$, substituting the saturated model gives

$$\log \frac{\theta_{11(1)}}{\theta_{11(2)}} = (\lambda^{XYZ}_{111} + \lambda^{XYZ}_{221} - \lambda^{XYZ}_{121} - \lambda^{XYZ}_{211}) - (\lambda^{XYZ}_{112} + \lambda^{XYZ}_{222} - \lambda^{XYZ}_{122} - \lambda^{XYZ}_{212}),$$

which equals λ^{XYZ}_{111} when second-category parameters are set to 0. When X, Y are binary and second-level parameters are zero, $\log \theta_{11(k)} = \lambda^{XY}_{11}$.

8.2.4 Alcohol, cigarette, and marijuana use example

Example 8.8 (Alcohol, cigarette, marijuana use)

1992 survey of 2276 high-school seniors near Dayton, Ohio: ever used alcohol (A), cigarettes (C), marijuana (M).

		Observed: M		Fit of (AC, AM, CM)	
A	C	yes	no	yes	no
yes	yes	911	538	910.4	538.6
yes	no	44	456	44.6	455.4
no	yes	3	43	3.6	42.4
no	no	2	279	1.4	279.6

Estimated odds ratios under several models:

	Conditional			Marginal		
Model	AC	AM	CM	AC	AM	CM
(A, C, M)	1.0	1.0	1.0	1.0	1.0	1.0
(AC, M)	17.7	1.0	1.0	17.7	1.0	1.0
(AM, CM)	1.0	61.9	25.1	2.7	61.9	25.1
(AC, AM, CM)	7.8	19.8	17.3	17.7	61.9	25.1
(ACM), M=yes	13.8	24.3	17.5	17.7	61.9	25.1
(ACM), M=no	7.7	13.5	9.7			

A term absent from a model gives a conditional odds ratio 1.0; e.g. under (AM, CM),

$$\frac{909.24 \times 0.24}{45.76 \times 4.76} = \frac{438.84 \times 179.84}{555.16 \times 142.16} = 1.0,$$

yet the fitted marginal AC odds ratio is 2.7. Under (AC, AM, CM) the AC conditional odds ratio is 7.8 at both levels of M, $= \exp(\hat{\lambda}_{11}^{AC} + \hat{\lambda}_{22}^{AC} - \hat{\lambda}_{12}^{AC} - \hat{\lambda}_{21}^{AC})$. Estimates depend heavily on the model.

8.3 Inference for loglinear models

Motivation. Fit, needed terms and association sizes are judged by the usual GLM deviance tests and intervals.

8.3.1 Chi-squared goodness-of-fit tests

$X^2 = \sum(n - \hat{\mu})^2/\hat{\mu}$ and $G^2 = 2\sum n \log(n/\hat{\mu})$ test fit, df = cells − nonredundant parameters. AIC $= -2$(maximized log likelihood − number of parameters) amounts to minimizing $G^2 - 2\,\text{df}$.

Example 8.9 (Goodness of fit for the student survey)

Model	G^2	X^2	df	P-value
(A, C, M)	1286.0	1411.4	4	< 0.001
(A, CM)	534.2	505.6	3	< 0.001
(C, AM)	939.6	824.2	3	< 0.001
(M, AC)	843.8	704.9	3	< 0.001
(AC, AM)	497.4	443.8	2	< 0.001
(AC, CM)	92.0	80.8	2	< 0.001
(AM, CM)	187.8	177.6	2	< 0.001
(AC, AM, CM)	0.4	0.4	1	0.54
(ACM)	0.0	0.0	0	–

Only the homogeneous association model fits; it also minimizes AIC.

8.3.2 Inference about conditional associations

Tests about a term compare nested models: $-2(L_0 - L_1) = G^2(M_0) - G^2(M_1)$. For XY conditional independence within (XY, XZ, YZ), $H_0 : \lambda_{ij}^{XY} = 0$ for all $(I-1)(J-1)$ parameters, with statistic

$$G^2[(XZ, YZ) \mid (XY, XZ, YZ)] = G^2(XZ, YZ) - G^2(XY, XZ, YZ), \qquad \text{df} = (I-1)(J-1).$$

This is a model-based alternative to the generalized CMH test.

Example 8.10 (Conditional associations in the student survey)

AC conditional independence: $G^2[(AM, CM) \mid (AC, AM, CM)] = 187.8 - 0.4 = 187.4$, df $= 1$, $P < 0.001$; AM and CM are likewise needed (91.6, 497.0). Baseline coding (last category $= 0$), (AC, AM, CM):

$$\hat{\lambda}_{11}^{AC} = 2.054 \text{ (SE} = 0.174), \quad \hat{\lambda}_{11}^{AM} = 2.986 \text{ (0.465)}, \quad \hat{\lambda}_{11}^{CM} = 2.848 \text{ (0.164)}.$$

These are conditional log odds ratios. 95% Wald intervals: AC $\exp(2.054 \pm 1.96 \times 0.174) =$ (5.5, 11.0), AM (8.0, 49.2), CM (12.5, 23.8). All are strong, the marginal ones stronger still.

With large samples, significance alone is not informative; confidence intervals show whether associations are practically important.

8.4 Loglinear models for higher dimensions

Motivation. The three-way ideas extend to more variables, at the cost of many possible terms and sparse tables.

8.4.1 Four-way contingency tables

For variables W, X, Y, Z, the model with all two-factor terms,

$$\begin{aligned}\log \mu_{hijk} = \lambda &+ \lambda_h^W + \lambda_i^X + \lambda_j^Y + \lambda_k^Z \\ &+ \lambda_{hi}^{WX} + \lambda_{hj}^{WY} + \lambda_{hk}^{WZ} + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ},\end{aligned}$$

symbol (WX, WY, WZ, XY, XZ, YZ), has each pair conditionally dependent with odds ratios constant over the other two variables; removing a two-factor term gives conditional independence of that pair. Under (WXY, WZ, XZ, YZ), the WX, WY, XY associations may vary over the third member of the triple, while all associations with Z are homogeneous. The saturated model adds all three-factor terms and λ^{WXYZ}.

Proposition 8.11 (Odds ratios in models without a variable in higher-order terms)

If a variable V appears in no term of order three or more, then for every other variable U the conditional UV odds ratios are the same at every combination of levels of the remaining variables, and they equal the exponentiated contrast of λ^{UV}.

Proof. [Supplement] Fix the other variables and form $\mu_{ab\cdot}\mu_{cd\cdot}/(\mu_{ad\cdot}\mu_{cb\cdot})$ (U levels a, c; V levels b, d). In the log, a term without V appears at $U = a$ once with each sign (in μ_{ab}, μ_{ad}), likewise at $U = c$, so it cancels; terms without U cancel the same way. By hypothesis the only term with both is λ^{UV}, free of the other variables, so the log odds ratio is $\lambda_{ab}^{UV} + \lambda_{cd}^{UV} - \lambda_{ad}^{UV} - \lambda_{cb}^{UV}$ at every level combination. ■

8.4.2 Automobile accident example

Example 8.12 (Automobile accidents and seat belts)

Maine, 1991: 68,694 auto-accident passengers by gender (G), location (L: urban, rural), seat-belt use (S), injury (I).

Gender	Location	Seat belt	Injury no	Injury yes
Female	Urban	No	7287	996
		Yes	11587	759
	Rural	No	3246	973
		Yes	6134	757
Male	Urban	No	10381	812
		Yes	10969	380
	Rural	No	6123	1084
		Yes	6693	513

Goodness of fit:

Model	G^2	df	P
(G, I, L, S)	2792.8	11	< 0.0001
(GI, GL, GS, IL, IS, LS)	23.4	5	< 0.001
(GIL, GIS, GLS, ILS)	1.3	1	0.25
(GIL, GS, IS, LS)	18.6	4	0.001
(GIS, GL, IL, LS)	22.8	4	< 0.001
(GLS, GI, IL, IS)	7.5	4	0.11
(ILS, GI, GL, GS)	20.6	4	< 0.001

Estimated conditional odds ratios:

Odds ratio	(GI, GL, GS, IL, IS, LS)	(GLS, GI, IL, IS)
GI	0.58	0.58
IL	2.13	2.13
IS	0.44	0.44
GL (S=no / yes)	1.23 / 1.23	1.33 / 1.17
GS (L=urban / rural)	0.63 / 0.63	0.66 / 0.58
LS (G=female / male)	1.09 / 1.09	1.17 / 1.03

Pairwise model: IS conditional log odds ratio -0.814 (SE 0.028), odds ratio interval $\exp(-0.814 \pm 1.96 \times 0.028) = (0.42, 0.47)$: seat belts more than halve the odds of injury in each gender–location group. Injury is more likely in rural accidents and for females; males have 0.63 times the odds of females of wearing belts. In (GLS, GI, IL, IS), I is in no three-factor term, so its odds ratios with G, L, S are homogeneous (Proposition 8.11); the GL, GS, LS odds ratios vary with the third variable, e.g. $0.66 = (7273.2 \times 10{,}959.2)/(11{,}632.6 \times 10{,}358.9)$ for urban accidents.

8.4.3 Large samples and statistical versus practical significance

(GLS, GI, IL, IS) fits better than the pairwise model ($23.4 - 7.5 = 15.9$, df $= 1$, $P = 0.0001$), yet the fitted GL, GS, LS odds ratios barely change across the third variable: the significance reflects the huge n. With large samples focus on estimation; the pairwise model is adequate for most purposes.

8.4.4 Dissimilarity index

Definition (Dissimilarity index)

For cell counts $n_i = np_i$ and fitted values $\hat{\mu}_i = n\hat{\pi}_i$,

$$\hat{\Delta} = \frac{\sum_i |n_i - \hat{\mu}_i|}{2n} = \frac{1}{2}\sum_i |p_i - \hat{\pi}_i|.$$

It estimates the population index $\Delta = \frac{1}{2}\sum |\pi_i - \pi_i^{\text{model}}|$.

Proposition 8.13 (Meaning of $\hat{\Delta}$)

If $\sum \hat{\mu}_i = n$ (true for any loglinear model with an intercept), then $0 \le \hat{\Delta} \le 1$, and $n\hat{\Delta}$ is the minimum number of observations that must be moved to other cells to make the observed table equal to the fitted one.

Proof. [Supplement]

Step 1. $\sum |p_i - \hat{\pi}_i| \le \sum p_i + \sum \hat{\pi}_i = 2$, so $\hat{\Delta} \le 1$; clearly $\hat{\Delta} \ge 0$.

Step 2. Let $d_i = n_i - \hat{\mu}_i$. Since $\sum d_i = 0$, $\sum d_i^+ = \sum d_i^- = \frac{1}{2}\sum |d_i| = n\hat{\Delta}$.

Step 3. One move changes $\sum d_i^+$ by at most 1, so at least $\sum d_i^+$ moves are needed; moving d_i^+ out of each surplus cell into the deficit cells (total $\sum d_i^-$, equal) suffices (fractional moves allowed). So the minimum is $n\hat{\Delta}$. ■

$\hat{\Delta} < 0.02$ or 0.03 means the model fits closely in practice even if rejected; $\hat{\Delta}$ overestimates a Δ near 0, especially for small n. Accident data: $\hat{\Delta} = 0.008$ (pairwise), 0.003 ((GLS, GI, IL, IS)): moving under 1% of the data gives a perfect fit.

8.5 Loglinear–logit model connection

Motivation. Every logit model with categorical predictors is a loglinear model containing the full interaction among the predictors.

8.5.1 Using logit models to interpret loglinear models

Proposition 8.14 (Logit form of (XY, XZ, YZ))

If Y is binary and (XY, XZ, YZ) holds, then

$$\text{logit}\, P(Y = 1 \mid X = i, Z = k) = \alpha + \beta_i^X + \beta_k^Z, \tag{8.8}$$

with $\alpha = \lambda_1^Y - \lambda_2^Y$, $\beta_i^X = \lambda_{i1}^{XY} - \lambda_{i2}^{XY}$, $\beta_k^Z = \lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}$.

Proof. The logit is $\log(\mu_{i1k}/\mu_{i2k})$. Subtracting (8.5) at $j = 2$ from (8.5) at $j = 1$, the terms $\lambda, \lambda_i^X, \lambda_k^Z, \lambda_{ik}^{XZ}$ cancel, leaving

$$(\lambda_1^Y - \lambda_2^Y) + (\lambda_{i1}^{XY} - \lambda_{i2}^{XY}) + (\lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}).$$

■

The λ^{XZ} terms, which describe association among explanatory variables, cancel: the logit model $(X + Z)$ says nothing about it.

8.5.2 Auto accident example revisited

Example 8.15 (Injury as a response)

For (GLS, GI, IL, IS), the logit on injury cancels all terms without I, including λ^{GLS}, giving the logit model $(G + L + S)$:

$$\operatorname{logit} P(I = 1 \mid G = g, L = l, S = s) = \alpha + \beta_g^G + \beta_l^L + \beta_s^S, \qquad \beta_s^S = \lambda_{1s}^{IS} - \lambda_{2s}^{IS}.$$

Fitted values, G^2, df and standardized residuals are identical. The seat-belt log odds ratio is $\hat\beta_1^S - \hat\beta_2^S = -0.817$ in the logit model and $\hat\lambda_{11}^{IS} + \hat\lambda_{22}^{IS} - \hat\lambda_{12}^{IS} - \hat\lambda_{21}^{IS} = -0.817$ in the loglinear one. The loglinear model treats the 16 counts as Poisson; the logit model fixes the GLS margin and treats $\{n_{g1ls}\}$ as binomial.

8.5.3 Correspondence between loglinear and logit models

The logit on Y in (XY, YZ) is also (8.8), but (XY, YZ) additionally forces $X \perp Z \mid Y$, so it is *not* equivalent: the loglinear model must contain the full interaction among the explanatory variables.

Theorem 8.16 (Loglinear–logit equivalence)

Let Y be the response and $\mathbf{x}$ index the combinations of the explanatory variables. Consider a loglinear model containing the full interaction term among the explanatory variables, $\log \mu_{\mathbf{x}j} = \gamma_{\mathbf{x}} + \eta_{\mathbf{x}j}(\boldsymbol{\beta})$, where $\gamma_{\mathbf{x}}$ is unrestricted and $\eta_{\mathbf{x}j}$ collects the terms involving Y (with $\eta_{\mathbf{x}J} = 0$). Then the ML fitted values under Poisson sampling coincide with those of the baseline-category logit model $\log(\pi_{j|\mathbf{x}}/\pi_{J|\mathbf{x}}) = \eta_{\mathbf{x}j}(\boldsymbol{\beta})$ fitted with the margins $n_{\mathbf{x}+}$ fixed, and the ML estimates of $\boldsymbol{\beta}$ agree.

Proof. [Supplement]

Step 1. Reparametrize: $\mu_{\mathbf{x}+} = e^{\gamma_{\mathbf{x}}} \sum_j e^{\eta_{\mathbf{x}j}}$ and $\pi_{j|\mathbf{x}} = e^{\eta_{\mathbf{x}j}} / \sum_h e^{\eta_{\mathbf{x}h}}$, so $\mu_{\mathbf{x}j} = \mu_{\mathbf{x}+}\pi_{j|\mathbf{x}}$. Since $\gamma_{\mathbf{x}}$ is free, $(\{\mu_{\mathbf{x}+}\}, \boldsymbol{\beta})$ ranges over $(0,\infty)^{\#\mathbf{x}} \times \{\text{all } \boldsymbol{\beta}\}$, a product set (variation independence).

Step 2. The Poisson log likelihood splits:

$$\sum_{\mathbf{x},j} (n_{\mathbf{x}j} \log \mu_{\mathbf{x}j} - \mu_{\mathbf{x}j}) = \sum_{\mathbf{x}} (n_{\mathbf{x}+} \log \mu_{\mathbf{x}+} - \mu_{\mathbf{x}+}) + \sum_{\mathbf{x},j} n_{\mathbf{x}j} \log \pi_{j|\mathbf{x}}(\boldsymbol{\beta}).$$

Step 3. The two parts depend on separate parameters, so they are maximized separately: $\hat\mu_{\mathbf{x}+} = n_{\mathbf{x}+}$, and $\hat{\boldsymbol{\beta}}$ maximizes the second part, which is exactly the product-multinomial (binomial if $J = 2$) log likelihood of the logit model. Hence $\hat\mu_{\mathbf{x}j} = n_{\mathbf{x}+}\hat\pi_{j|\mathbf{x}}$ in both models. ■

Loglinear symbol	Logit model	Logit symbol
(Y, XZ)	α	$(-)$
(XY, XZ)	$\alpha + \beta_i^X$	(X)
(YZ, XZ)	$\alpha + \beta_k^Z$	(Z)
(XY, YZ, XZ)	$\alpha + \beta_i^X + \beta_k^Z$	$(X + Z)$
(XYZ)	$\alpha + \beta_i^X + \beta_k^Z + \beta_{ik}^{XZ}$	$(X*Z)$

With several responses the loglinear approach is more general; with one, logit models are more natural.

8.5.4 Generalized loglinear model

Write the counts as a vector $\mathbf{n} = (n_1, \ldots, n_N)'$ with means $\boldsymbol{\mu}$. An ordinary loglinear model is

$$\log \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}. \tag{8.9}$$

For instance, independence in a 2×2 table with $\lambda_2^X = \lambda_2^Y = 0$ is

$$\begin{pmatrix} \log \mu_{11} \\ \log \mu_{12} \\ \log \mu_{21} \\ \log \mu_{22} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ \lambda_1^X \\ \lambda_1^Y \end{pmatrix}.$$

The *generalized loglinear model* is $\mathbf{C}\log(\mathbf{A}\boldsymbol{\mu}) = \mathbf{X}\boldsymbol{\beta}$. With $\mathbf{A} = \mathbf{I}_4$, $\mathbf{C} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$, $\mathbf{X} = (1, 1)'$, $\boldsymbol{\beta} = \alpha$, both row logits equal α: independence. The class includes logit models and marginal models.

8.6 Loglinear model fitting: likelihood equations

Motivation. Fitted values match marginal tables because those margins are the sufficient statistics of a Poisson exponential family.

Derivations use Poisson sampling, which needs no constraint on parameters; Theorem 8.21 transfers them to multinomial sampling.

8.6.1 Minimal sufficient statistics

The kernel of the Poisson log likelihood for a three-way table is

$$L(\boldsymbol{\mu}) = \sum_{i,j,k} n_{ijk} \log \mu_{ijk} - \sum_{i,j,k} \mu_{ijk}. \tag{8.10}$$

Substituting the saturated model,

$$\begin{aligned} L = n\lambda + \sum_i n_{i++}\lambda_i^X + \sum_j n_{+j+}\lambda_j^Y + \sum_k n_{++k}\lambda_k^Z + \sum_{i,j} n_{ij+}\lambda_{ij}^{XY} + \sum_{i,k} n_{i+k}\lambda_{ik}^{XZ} \\ + \sum_{j,k} n_{+jk}\lambda_{jk}^{YZ} + \sum_{i,j,k} n_{ijk}\lambda_{ijk}^{XYZ} - \sum_{i,j,k} \exp(\lambda + \lambda_i^X + \cdots + \lambda_{ijk}^{XYZ}). \end{aligned} \tag{8.11}$$

Theorem 8.17 (Sufficient statistics are marginal tables)

For a hierarchical loglinear model, the minimal sufficient statistics are the marginal tables corresponding to the terms in the model symbol: e.g. $\{n_{i++}\}, \{n_{+j+}\}, \{n_{++k}\}$ for (X, Y, Z); $\{n_{ij+}\}, \{n_{++k}\}$ for (XY, Z); $\{n_{ij+}\}, \{n_{+jk}\}$ for (XY, YZ); $\{n_{ij+}\}, \{n_{i+k}\}, \{n_{+jk}\}$ for (XY, XZ, YZ).

Proof.

Step 1. With absent parameters zero in (8.11), the family is exponential with canonical statistics the marginal tables of the terms present; they are sufficient (factorization) and minimal (full rank after removing redundant parameters).

Step 2. A lower-order margin is a sum of a higher-order one (e.g. $n_{i++} = \sum_j n_{ij+}$), so only the margins of the highest-order terms (the generators) are needed. ■

For the saturated model the full table is sufficient: no data reduction.

8.6.2 Likelihood equations for loglinear models

In the matrix form (8.9), $\log \mu_i = \sum_j x_{ij}\beta_j$ and

$$L(\boldsymbol{\beta}) = \sum_i n_i\Big(\sum_j x_{ij}\beta_j\Big) - \sum_i \exp\Big(\sum_j x_{ij}\beta_j\Big). \tag{8.12}$$

Theorem 8.18 (Likelihood equations match margins)

(a) The likelihood equations are

$$\mathbf{X}'\mathbf{n} = \mathbf{X}'\hat{\boldsymbol{\mu}}\,: \tag{8.13}$$

sufficient statistics equal their fitted (expected) values.
(b) For a hierarchical model, (8.13) is equivalent to equality of the observed and fitted marginal tables in the model symbol. For (XZ, YZ):

$$\hat{\mu}_{i+k} = n_{i+k} \text{ for all } i,k, \qquad \hat{\mu}_{+jk} = n_{+jk} \text{ for all } j,k. \tag{8.14}$$

Proof.

Step 1. Since $\partial \exp(\sum_j x_{ij}\beta_j)/\partial\beta_j = x_{ij}\mu_i$,

$$\frac{\partial L}{\partial \beta_j} = \sum_i n_i x_{ij} - \sum_i \mu_i x_{ij}, \qquad j = 1,\ldots,p.$$

Setting these to 0 gives (8.13).

Step 2. (8.13) says $\mathbf{v}'\mathbf{n} = \mathbf{v}'\hat{\boldsymbol{\mu}}$ for all $\mathbf{v} \in \mathrm{col}(\mathbf{X})$. By Lemma 8.5 this space consists of the functions $\sum_r g_r(\mathbf{x}_{C_r})$, spanned by the indicators $\mathbf{v} = \mathbb{1}\{\mathbf{x}_{C_r} = \mathbf{c}\}$, for which $\mathbf{v}'\mathbf{n}$ is the marginal count at $\mathbf{c}$. So (8.13) is equality of all generator margins; for (XZ, YZ) this is (8.14). ■

So fitted values agree with the data on certain margins while their associations follow the model.

8.6.3 Birch's results for loglinear models

Theorem 8.19 (Birch)

There is at most one $\hat{\boldsymbol{\mu}}$ that both satisfies the loglinear model and matches the data in the minimal sufficient statistics, and if it exists it is the ML estimate. Conversely, any ML estimate has these two properties.

Proof. [Supplement]

Step 1. Write $\boldsymbol{\eta} = \log \boldsymbol{\mu} \in \mathcal{V} = \mathrm{col}(\mathbf{X})$. As a function of $\boldsymbol{\eta}$, $L = \mathbf{n}'\boldsymbol{\eta} - \sum_i e^{\eta_i}$ is strictly concave on $\mathbb{R}^N$: its Hessian is $-\mathrm{diag}(e^{\eta_i})$, negative definite. Its restriction to the subspace $\mathcal{V}$ is still strictly concave.

Step 2. A point $\boldsymbol{\eta} \in \mathcal{V}$ is stationary for $L|_{\mathcal{V}}$ iff the gradient $\mathbf{n} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{V}$, i.e. $\mathbf{X}'(\mathbf{n} - \boldsymbol{\mu}) = \mathbf{0}$, which is margin matching by Theorem 8.18.

Step 3. A stationary point of a differentiable concave function is a global maximum, unique under strict concavity. So a model $\boldsymbol{\mu}$ matching the margins is the unique MLE; conversely the (interior) MLE is stationary, hence matches the margins. ■

E.g. for two-way independence, $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$ satisfies the model and the margins, so it is the MLE.

8.6.4 Direct versus iterative calculation of fitted values

Theorem 8.20 (Closed-form (direct) estimates)

The ML fitted values for three-way models other than (XY, XZ, YZ) are:

Model	Probabilistic form	Fitted value $\hat{\mu}_{ijk}$
(X, Y, Z)	$\pi_{i++}\pi_{+j+}\pi_{++k}$	$n_{i++}n_{+j+}n_{++k}/n^2$
(XY, Z)	$\pi_{ij+}\pi_{++k}$	$n_{ij+}n_{++k}/n$
(XY, XZ)	$\pi_{ij+}\pi_{i+k}/\pi_{i++}$	$n_{ij+}n_{i+k}/n_{i++}$
(XY, XZ, YZ)	$\psi_{ij}\phi_{jk}\omega_{ik}$	iterative (IPF)
(XYZ)	none	n_{ijk}

(other models by symmetry, e.g. (XZ, YZ): $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$).

Proof. Take (XZ, YZ); the others are identical in structure.

Step 1. By Theorem 8.6, under the model $\mu_{ijk} = \mu_{i+k}\mu_{+jk}/\mu_{++k}$.

Step 2. Define $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$. It has the product form $A_{ik}B_{jk}$, so it satisfies the model.

Step 3. Summing over j: $\hat{\mu}_{i+k} = n_{i+k}n_{++k}/n_{++k} = n_{i+k}$; summing over i: $\hat{\mu}_{+jk} = n_{+jk}$. Thus it matches the sufficient margins (8.14).

Step 4. By Birch's theorem it is the MLE. For (X, Y, Z), (XY, Z), (XY, XZ) the displayed fits likewise have the factorized form and reproduce the sufficient margins (e.g. for (XY, XZ): $\sum_k n_{ij+}n_{i+k}/n_{i++} = n_{ij+}$ and $\sum_j n_{ij+}n_{i+k}/n_{i++} = n_{i+k}$). For (XYZ), $\hat{\mu} = \mathbf{n}$ trivially. ■

Models with closed-form estimates are *direct* (*decomposable*: products and ratios of fitted margins). Unsaturated models with all two-factor terms, such as (XY, XZ, YZ), are not; software iterates anyway.

8.6.5 Connection between multinomial and Poisson loglinear models

Theorem 8.21 (Poisson and multinomial give the same estimates)

Write the Poisson model as $\log \mu_i = \alpha + \mathbf{x}_i'\boldsymbol{\beta}$ (α the intercept). Then the Poisson log likelihood decomposes as

$$L(\alpha, \boldsymbol{\beta}) = \Big[\sum_i n_i \log \pi_i(\boldsymbol{\beta})\Big] + [n \log \tau - \tau], \qquad \pi_i(\boldsymbol{\beta}) = \frac{e^{\mathbf{x}_i'\boldsymbol{\beta}}}{\sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}}, \tag{8.15}$$

where $\tau = \sum_i \mu_i$. The first bracket is the multinomial log likelihood given n, the second the Poisson log likelihood of n. Consequently $\hat{\boldsymbol{\beta}}$ and its asymptotic covariance are the same under Poisson and multinomial sampling.

Proof.

Step 1. $L = \sum n_i \log \mu_i - \sum \mu_i = \sum n_i(\alpha + \mathbf{x}_i'\boldsymbol{\beta}) - \tau = n\alpha + \sum n_i \mathbf{x}_i'\boldsymbol{\beta} - \tau$.

Step 2. Since $\tau = e^{\alpha}\sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}$, $\alpha = \log \tau - \log \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}$. Substituting,

$$L = \sum_i n_i \mathbf{x}_i'\boldsymbol{\beta} - n \log \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}} + n \log \tau - \tau = \sum_i n_i \log \pi_i(\boldsymbol{\beta}) + (n \log \tau - \tau).$$

Step 3. The map $(\alpha, \boldsymbol{\beta}) \mapsto (\tau, \boldsymbol{\beta})$ is one-to-one onto $(0, \infty) \times \mathbb{R}^p$, and $\boldsymbol{\beta}$ enters only the first bracket. Hence $\hat{\boldsymbol{\beta}}$ maximizes the multinomial log likelihood, $\hat{\tau} = n$, and the information matrix is block

diagonal in $(\tau, \boldsymbol{\beta})$, so the $\boldsymbol{\beta}$ block of the inverse information is the same for both sampling schemes. ■

The same holds for several independent multinomial samples if the model contains the term of the fixed margin (e.g. λ^{XZ} for fixed $\{n_{i+k}\}$), forcing $\hat{\mu}_{i+k} = n_{i+k}$ (Theorem 8.16). For one multinomial sample, $\widehat{\text{cov}}(\hat{\boldsymbol{\beta}}) = \{\mathbf{X}'[\text{diag}(\hat{\boldsymbol{\mu}}) - \hat{\boldsymbol{\mu}}\hat{\boldsymbol{\mu}}'/n]\mathbf{X}\}^{-1}$ ($\mathbf{X}$ without intercept), the Poisson covariance $\{\mathbf{X}' \text{diag}(\hat{\boldsymbol{\mu}})\mathbf{X}\}^{-1}$ with the α row and column deleted.